Layer 10: Marcus-Theory Electron Transfer

MB Redox Kinetics: From Tether Distance to SWV Rate Constant

Eton iGEM 2026, Dry Lab | Layer 10 of 17-Layer Biosensor Pipeline | `electron_transfer.py`

Layer 10 derives the **standard electron-transfer rate constant** k^0_{ET} for the methylene-blue (MB) redox label tethered to the gold electrode surface. k^0_{ET} governs the peak current, peak shape, and optimal scan frequency in the SWV measurement (Layer 11). Its value depends on three physically distinct quantities: the **tunnelling distance** d between MB and the gold surface (set by the probe conformation from Layer 8), the **reorganisation energy** λ, and the **electrode potential** E relative to the MB formal potential $E^{\circ\prime}_{\mathrm{MB}}(\mathrm{pH})$.

The model proceeds as follows. Section I derives the rate constant from the Marcus (1965) [1] electrode reaction framework. Section II quantifies the distance dependence using the tunnelling decay coefficient β measured by Smalley et al. (1995) [2]. Section III estimates λ from the Chidsey (1991) [3] Fc/Au measurements as the closest available analogue. Section IV derives the Butler-Volmer (BV) approximation; Section V derives the full Chidsey-Marcus integral valid at large overpotentials. Sections VI–VIII add the pH correction, temperature correction, and double-layer (Frumkin) correction. Section IX synthesises all contributions into $k^0_{\mathrm{ET,eff}}$. Section X states limitations.

I. The Marcus Electrode Rate Equation

Electrochemical activation energy

Marcus (1965) [1] derived the heterogeneous electron-transfer rate constant for an electrode reaction:

$$\mathrm{Ox} + ne^- \rightleftharpoons \mathrm{Red} \tag{I.1}$$

The fundamental result (Marcus 1965, Eq. 33, with $\kappa \approx 1$, $\rho \approx 1$) is:

$$k_{\mathrm{het}} = Z_{\mathrm{het}} \exp\!\left(-\frac{\Delta F^*}{k_{\mathrm{B}}T}\right) \tag{I.2}$$

where $Z_{\mathrm{het}} = (k_{\mathrm{B}}T/2\pi m^*)^{1/2} \approx 10^4\ \mathrm{cm\,s^{-1}}$ is the heterogeneous collision frequency (Marcus 1965, Eq. 34), m^* is the reduced mass of the reactant, and ΔF^* is the free energy of activation.

For an electrode reaction with formal overpotential $\eta = E - E^{\circ\prime}$ (work terms $w^r \approx w^p \approx 0$ for the neutral/uncharged MB surface species), Marcus 1965 Eq. 82 gives:

$$\Delta F^* = \frac{\lambda}{4} + \frac{ne(E - E^{\circ\prime})}{2} + \frac{\left[ne(E - E^{\circ\prime})\right]^2}{4\lambda} \tag{I.3}$$

At $E = E^{\circ\prime}$ the overpotential is zero, and Eq. (I.3) reduces to:

$$\Delta F^*_0 = \frac{\lambda}{4} \tag{I.4}$$

giving the standard rate constant:

$$k^0_{\mathrm{ET}} = Z_{\mathrm{het}} \exp\!\left(-\frac{\lambda}{4k_{\mathrm{B}}T}\right) \tag{I.5}$$

Transfer coefficient

Differentiating ΔF^* in Eq. (I.3) with respect to $ne(E - E^{\circ\prime})$ at $E = E^{\circ\prime}$ (Marcus 1965, Eq. 87b with work terms neglected):

$$\alpha = \frac{1}{2} + \frac{ne(E - E^{\circ\prime})}{2\lambda} \tag{I.6}$$

At $E = E^{\circ\prime}$, $\alpha = 0.5$, as expected for a symmetric potential energy surface intersection. Equation (I.6) predicts deviations from 0.5 only when the overpotential becomes comparable to $\lambda/2$ (the Marcus inverted region boundary).

Reorganisation energy additivity

λ is the total reorganisation energy (Marcus 1965, Eq. 83):

$$\lambda = \lambda_i + \lambda_o \tag{I.7}$$

where λ_i is the *inner-sphere* contribution (bond length and angle changes within MB upon reduction) and λ_o is the *outer-sphere* (solvent/dielectric) contribution. For the electrode geometry, Marcus 1965 Eq. 90 gives λ_o in the dielectric continuum approximation:

$$\lambda_o = \frac{(ne)^2}{2}\left(\frac{1}{2a_1} - \frac{1}{R}\right)\left(\frac{1}{D_{\text{op}}} - \frac{1}{D_s}\right) \tag{I.8}$$

where a_1 is the effective radius of the reactant (MB, $a_1 \approx 4$ Å), R is twice the distance from the MB centre to the electrode surface, $D_{\text{op}} \approx 1.78$ is the optical (high-frequency) dielectric constant of water, and $D_s \approx 78$ is the static dielectric constant.

II. Distance Dependence: The Tunnelling Decay Constant

Exponential distance law

The electronic coupling between the MB redox centre and the gold electrode decays exponentially with the tunnelling distance d. Smalley et al. (1995) [2] characterised this for ferrocene-terminated alkanethiol monolayers on gold using the indirect laser-induced temperature jump (ILIT) method, which accesses rates too fast for conventional chronoamperometry ($\leq 10^4$ s^{-1}). For $5 \leq n \leq 9$ methylene units, they report (Smalley 1995, abstract and text):

$$k_{T,s}(n) = k_{T,s,n=0}\,\exp(-\beta_n\, n) \tag{II.1}$$

with confirmed values:

$$k_{T,s,n=0} = 6.0 \times 10^8 \text{ s}^{-1} \tag{II.2}$$

$$\beta_n = 1.21 \pm 0.05 \text{ per CH}_2 \tag{II.3}$$

Conversion to a per-angstrom decay constant

Each methylene unit contributes $d_{\text{CH}_2} \approx 1.27$ Å to the all-*trans* chain length. Converting Eq. (II.3):

$$\beta = \frac{\beta_n}{d_{\text{CH}_2}} = \frac{1.21}{1.27 \text{ Å}} \approx 0.95 \text{ Å}^{-1} \tag{II.4}$$

so that in terms of physical distance (Smalley 1995, Eq. 2):

$$k^0(d) = k_{T,s,n=0}\,\exp(-\beta\, d) = 6.0 \times 10^8\,\exp(-0.95\, d[\text{Å}])\text{ s}^{-1} \tag{II.5}$$

Numerical evaluation across the clinical tether range

The effective tunnelling distance d is set by the probe conformation distribution $P(d)$ from Layer 8. Pending that measurement, Table 1 evaluates Eq. (II.5) across a physically plausible range.

d (nm)	d (Å)	k^0 (s^{-1}, Eq. II.5)
1.0	10	$\sim 4.5 \times 10^4$
1.5	15	~ 430
2.0	20	~ 4.0
2.5	25	~ 0.037

Table 1: Standard ET rate constant as a function of tunnelling distance, from Eq. (II.5) (Smalley 1995 [2] parameters). The tether peptide contour length is $L_c \approx 5$ nm; d_{eff} is shorter because of chain flexibility (Layer 8). Values span several orders of magnitude: the SWV-accessible range is approximately 1–10^4 s^{-1} (Section IX).

Caveat. Smalley 1995 used rigid ferrocene-terminated *alkyl* chains. Our probe uses a PEG-peptide tether, for which β_{PEG} has not been measured in this system. Alkyl chains are the closest well-characterised analogue; the value $\beta \approx 0.95$ Å^{-1} should be treated as an estimate until k^0 is measured empirically on the fabricated electrode (Limitation 1, Section X).

III. Reorganisation Energy

Reference measurement: Chidsey (1991)

Chidsey (1991) [3] measured the rate constant and its full energy dependence for ferrocene-terminated alkanethiol monolayers on polycrystalline gold in 1 M $HClO_4$, over a 46 °C temperature range and a 1.6 V potential window.

The best-fit reorganisation energy from the full Marcus-Chidsey integral (Section V) is confirmed (Chidsey 1991, abstract and Fig. 4):

$$\lambda_{\mathrm{Fc/Au}} = 0.85 \text{ eV} \tag{III.1}$$

The electronic coupling prefactor (tunnelling attempt frequency) is (Chidsey 1991, abstract):

$$\nu_p = 6.73 \times 10^4 \text{ s}^{-1}\text{eV}^{-1} \tag{III.2}$$

The standard rate constant for $n = 16$ (C16 chain, $d \approx 20$ Å) is confirmed (Chidsey 1991, p. 920):

$$k^0_{\mathrm{Fc,\ C16}} = 1.3 \text{ s}^{-1} \tag{III.3}$$

Application to MB

MB undergoes a two-electron, two-proton reduction:

$$\mathrm{MB}^+ + 2\mathrm{H}^+ + 2e^- \rightleftharpoons \mathrm{LMB} \tag{III.4}$$

The outer-sphere λ_o depends primarily on the gold-electrode geometry and the aqueous medium, which are the same as in Chidsey's system. The inner-sphere λ_i for MB is not characterised in any paper in `papers/layer_10/`. We therefore apply $\lambda \approx 0.85$ eV as the best available estimate, with the caveat that λ_i^{MB} may add 0.05–0.15 eV (Limitation 2).

Outer-sphere estimate from Marcus continuum theory

As a consistency check, Eq. (I.8) with $n = 1$ (treating MB reduction as equivalent to a one-electron process at the outer-sphere level, since λ_o depends on the charge transferred per step), $a_1 = 4$ Å, $R \approx 2d = 20$ Å, $D_{\mathrm{op}} = 1.78$, $D_s = 78$:

$$\begin{aligned}
\lambda_o &= \frac{(1.6 \times 10^{-19})^2}{2}\left(\frac{1}{2 \times 4 \times 10^{-10}} - \frac{1}{20 \times 10^{-10}}\right)\left(\frac{1}{1.78} - \frac{1}{78}\right) \\
&= \frac{2.56 \times 10^{-38}}{2} \text{ C}^2 \times \left(1.25 \times 10^9 - 5.0 \times 10^8\right) \text{ m}^{-1} \times 0.549 \text{ (F/m)}^{-1} \\
&\approx 1.28 \times 10^{-38} \times 7.5 \times 10^8 \times 0.549/(8.854 \times 10^{-12}) \text{ J} \\
&\approx 0.60 \text{ eV}
\end{aligned}$$

This is consistent with the Chidsey measurement $\lambda = 0.85$ eV, since λ_i adds to λ_o in Eq. (I.7).

IV. Butler-Volmer Approximation

Derivation from the Marcus activation energy

For $|ne(E - E^{\circ\prime})| \ll \lambda$ (moderate overpotentials), Taylor-expanding Eq. (I.3) to first order in $\eta = E - E^{\circ\prime}$:

$$\Delta F^* \approx \frac{\lambda}{4} + \frac{ne\eta}{2} \tag{IV.1}$$

Substituting into Eq. (I.2) and separating the cathodic (k_{f}) and anodic (k_{b}) contributions:

$$k_{\mathrm{f}}(\eta) = k^0_{\mathrm{ET}} \exp\left(-\frac{\alpha F \eta}{RT}\right) \tag{IV.2}$$

$$k_{\mathrm{b}}(\eta) = k^0_{\mathrm{ET}} \exp\left(\frac{(1-\alpha) F \eta}{RT}\right) \tag{IV.3}$$

with $\alpha = 0.5$ at $E = E^{\circ\prime}$, $F = 96{,}485$ C mol^{-1}, $R = 8.314$ J mol^{-1} K^{-1}. This is the Butler-Volmer expression confirmed in Chidsey 1991, Eq. 3 [3].

Validity range

The BV approximation is valid when $|e\eta|/\lambda \ll 1$. For $\lambda = 0.85$ eV, this requires $|\eta| \ll 425$ mV. The SWV scan window for MB extends from -0.65 V to $+0.1$ V vs Ag/AgCl, giving a maximum overpotential $|\eta|_{\max} = 650 - 290 = 360$ mV < 425 mV; BV is marginally valid across most of the scan, but the Marcus correction at the negative scan limit ($\eta \approx -0.36$ V) is not negligible. The full Chidsey-Marcus integral (Section V) removes this limitation.

Numerical evaluation at T = 37°C

At $T = 310$ K, $F/RT = 37.51$ V^{-1}. For $\eta = -0.1$ V (100 mV cathodic):

$$k_{\mathrm{f}}(-0.1 \text{ V}) = k^0_{\mathrm{ET}} \exp(0.5 \times 37.51 \times 0.1) = k^0_{\mathrm{ET}} \times 6.55 \tag{IV.4}$$

For $\eta = -0.3$ V (at the cathodic scan limit near the MB peak):

$$k_{\mathrm{f}}(-0.3 \text{ V}) = k^0_{\mathrm{ET}} \exp(0.5 \times 37.51 \times 0.3) = k^0_{\mathrm{ET}} \times 282 \tag{IV.5}$$

V. Full Chidsey-Marcus Integral

Electronic density of states of the metal

The Butler-Volmer model treats the metal electrode as having a single energy level at the Fermi energy. The Marcus-Chidsey model (Chidsey 1991 [3], Eqs. 4–8) correctly accounts for the continuum of metallic electronic states, weighted by the Fermi-Dirac distribution:

$$f_{\pm}(\varepsilon_i) = \frac{1}{1 + \exp\!\left[\pm e(\varepsilon_i - \bar{\mu})/(k_{\mathrm{B}}T)\right]} \tag{V.1}$$

The cathodic (+) and anodic (−) rate constants are (Chidsey 1991, Eqs. 7–8):

$$k_{\mathrm{f}}(E,T) = \nu_p\, P_n \int_{-\infty}^{+\infty} \exp\!\left[-\frac{(x - (\lambda^* + E^*))^2}{4\lambda^*}\right] \frac{1}{1+e^{x}}\, dx \tag{V.2}$$

$$k_{\mathrm{b}}(E,T) = \nu_p\, P_n \int_{-\infty}^{+\infty} \exp\!\left[-\frac{(x - (\lambda^* - E^*))^2}{4\lambda^*}\right] \frac{1}{1+e^{-x}}\, dx \tag{V.3}$$

where $\lambda^* = \lambda/(k_{\mathrm{B}}T)$, $E^* = e\eta/(k_{\mathrm{B}}T)$, and $P_n = 2\pi^{3/2}|V_n|^2\rho_m/h$ is a product of the electronic coupling $|V_n|$ and the metallic density of states ρ_m (Chidsey 1991, Eq. 5).

Key features of the integral

At **zero overpotential** ($E^* = 0$), both expressions give the same k^0_{ET}, recovering Eq. (I.5).

At **large cathodic overpotential** ($E^* \gg \lambda^*$), k_{f} plateaus at $\nu_p P_n \sqrt{\pi\lambda^*}$: the Marcus inverted region is suppressed because the Fermi-Dirac distribution provides states at all energies. This *plateau* is observed experimentally in Chidsey 1991 (Fig. 4A), and is absent in the BV model.

At **moderate overpotentials** ($|E^*| \ll \lambda^*$), the integrals reduce to the BV expressions in Eqs. (IV.2)–(IV.3).

VI. Formal Potential of Methylene Blue: pH Dependence

Nernst equation for the $2e^-/2H^+$ couple

The MB reduction is (Eq. III.4): two electrons and two protons are consumed. Applying the Nernst equation to this half-reaction:

$$E = E^{\circ\prime}_{\mathrm{MB}} - \frac{RT}{nF} \ln \frac{[\mathrm{LMB}]}{[\mathrm{MB}^+][\mathrm{H}^+]^2} \tag{VI.1}$$

At equilibrium in acidic media, setting activity quotient to unity and expressing $[\mathrm{H}^+] = 10^{-\mathrm{pH}}$:

$$E^{\circ\prime}_{\mathrm{MB}}(\mathrm{pH}) = E^{\circ\prime}_{\mathrm{MB}}(\mathrm{pH}_0) - \frac{n_H}{n_e} \cdot \frac{RT \ln 10}{F} \cdot (\mathrm{pH} - \mathrm{pH}_0) \tag{VI.2}$$

With $n_H = n_e = 2$, the ratio $n_H/n_e = 1$, so the coefficient is exactly 59.2 mV per pH unit at 25 °C and 61.6 mV per pH unit at 37 °C.

Confirmed reference value

Pheeney and Barton (2012) [4] measured the midpoint potential of MB covalently tethered to DNA on gold electrodes in aqueous buffer (pH 7). The confirmed value (Pheeney 2012, p. 7065, Fig. 2 and text):

$$E^{\circ\prime}_{\mathrm{MB}}(\mathrm{pH}\ 7) = -290\ \mathrm{mV\ vs\ Ag/AgCl} \tag{VI.3}$$

This is used as the reference in Eq. (VI.2):

$$E^{\circ\prime}_{\mathrm{MB}}(\mathrm{pH}) = -290 - 61.6\,(\mathrm{pH} - 7)\ \mathrm{mV\ vs\ Ag/AgCl} \tag{VI.4}$$

(using the 37 °C slope; the difference from the 25 °C value of 59.2 mV/pH is 4%).

Wound pH range

Wound fluid pH spans 6.5–8.5 across wound types (chronic wounds are slightly acidic). Applying Eq. (VI.4):

pH	$E^{\circ\prime}_{\mathrm{MB}}$ (mV vs Ag/AgCl)	Shift from pH 7
6.5	$-290 + 31 = -259$	+31 mV
7.0	-290	reference
7.5	$-290 - 31 = -321$	−31 mV
8.0	$-290 - 62 = -352$	−62 mV

Table 2: MB formal potential across the wound pH range, from Eq. (VI.4) and Pheeney 2012 [4]. The ±62 mV shift across the pH 6.5–8.0 range is well within the MB peak FWHM of ≈ 33 mV (Layer 11); the peak fitting in Layer 12 must allow a floating peak position.

VII. Temperature Correction to k^0

Derivation from the Marcus activation energy

From Eq. (I.5), $k^0_{\mathrm{ET}} \propto \exp(-\lambda/4k_{\mathrm{B}}T)$. This gives an Arrhenius activation energy:

$$E_a = \frac{\lambda}{4} = \frac{0.85}{4} = 0.21 \text{ eV} \approx 20.3 \text{ kJ mol}^{-1} \tag{VII.1}$$

confirmed for the Fc/Au system by the Arrhenius plots in Chidsey 1991 (Fig. 4B, temperature range 1–47 °C). The temperature correction factor from $T_{\mathrm{ref}} = 298$ K to $T = 310$ K is:

$$\frac{k^0_{\mathrm{ET}}(310)}{k^0_{\mathrm{ET}}(298)} = \exp\left[\frac{\lambda}{4k_{\mathrm{B}}}\left(\frac{1}{298} - \frac{1}{310}\right)\right] = \exp\left[2{,}466\,\mathrm{K} \times 1.30 \times 10^{-4}\,\mathrm{K}^{-1}\right] = e^{0.320} \approx 1.38 \tag{VII.2}$$

The rate constant increases by approximately 38% at wound temperature (37 °C) relative to the 25 °C laboratory reference value.

Impact on SWV operating point

Because both the MB and Fc (reference) electrodes are on the same chip at the same temperature, the temperature shift of k^0_{ET} is identical for all probes. The ratiometric signal $F_k(T)$ in Layer 13 is thus insensitive to uniform temperature changes to first order. However, if $k^0_{\mathrm{ET}}(T)$ crosses a regime boundary (e.g., from quasi-reversible to reversible as temperature rises), the peak height can change non-linearly. This is mitigated by pre-characterising the optimal f_{SWV} at wound temperature (37 °C, not 25 °C).

VIII. Double-Layer (Frumkin) Correction

Diffuse-layer potential at the SAM-coated electrode

Becka and Miller (1993) [5] showed that self-assembled ω-hydroxy thiol monolayers on gold dramatically reduce the diffuse-layer potential ϕ_2 relative to bare gold, because the hydrocarbon film drops nearly all of the applied potential *inside* the compact film. The double layer is modelled as two capacitors in series: the monolayer compact capacitor C_m and the Gouy-Chapman diffuse-layer capacitor C_{gc} [5], Eqs. 1–4). The effective potential at the outer Helmholtz plane (OHP) is:

$$\phi_2 = \frac{C_m}{C_{gc} + C_m} V_{\mathrm{app}} \tag{VIII.1}$$

For our PEG-MCH mixed SAM with a charged peptide probe, ϕ_2 is further reduced by the high ionic strength of wound fluid ($I \approx 150$ mM), which compresses the Debye layer to $\kappa^{-1} \approx 0.77$ nm. At this ionic strength, the Gouy-Chapman diffuse-layer capacitance $C_{gc} = 238.2c^{1/2}$ μF cm^{-2} (Becka 1993, Eq. 7; c in mol/L) gives:

$$C_{gc}(150 \text{ mM}) = 238.2 \times 0.15^{1/2} \approx 92 \ \mu\text{F cm}^{-2} \tag{VIII.2}$$

For a typical monolayer capacitance $C_m \approx 2$ μF cm^{-2}, the ratio $C_m/(C_{gc} + C_m) \approx 0.021$, so ϕ_2 is only 2% of the applied potential. The Frumkin correction (Becka 1993, Eq. 5):

$$k_{\mathrm{corr}}(E) = k(E)\, \exp\left(-\frac{z\, e\, \phi_2(E)}{k_{\mathrm{B}}T}\right) \tag{VIII.3}$$

(where z is the charge of the electroactive species at the OHP) reduces to a $< 5\%$ correction for surface-confined MB ($z \approx 0$ near $E^{\circ\prime}_{\mathrm{MB}}$, since MB^+ is neutral at its formal potential). This correction is included in the model but is second-order.

IX. Synthesis: Effective $k^0_{\mathbf{ET}}$

Combining all contributions, the effective standard ET rate constant passed to the SWV forward model (Layer 11) is:

$$k^0_{\mathrm{ET,eff}} = \underbrace{k^0_{\mathrm{ET}}(d)}_{\text{Section II}} \times \underbrace{f_T}_{\text{Eq. VII.2}} \times \underbrace{f_{\phi_2}(E)}_{\text{Eq. VIII.3}} \tag{IX.1}$$

where $k^0_{\mathrm{ET}}(d)$ is from Eq. (II.5) evaluated at the mean effective distance $\langle d \rangle$ from Layer 8, and $f_T = 1.38$ for wound-temperature operation. The potential-dependent factor $f_{\phi_2}(E)$ is incorporated into the BV/Chidsey-Marcus rate expressions directly.

At the expected clinically relevant tether distances, the estimated $k^0_{\mathrm{ET,eff}}$ range is:

$\langle d \rangle$ (nm)	k^0_{ET} at 25°C (s^{-1})	$f_T \times k^0_{\mathrm{ET}}$ at 37°C (s^{-1})	SWV regime (Section X)
1.0	$\sim 4.5 \times 10^4$	$\sim 6.2 \times 10^4$	reversible
1.5	~ 430	~ 590	quasi-reversible
2.0	~ 4.0	~ 5.5	quasi-reversible
2.5	~ 0.04	~ 0.05	irreversible

Table 3: Estimated $k^0_{\mathrm{ET,eff}}$ from Eq. (IX.1) using $\beta = 0.95$ Å^{-1} (Smalley 1995 [2]) and $f_T = 1.38$ (Chidsey 1991 [3], Eq. VII.2). The SWV regime column assumes $f_{\mathrm{SWV}} = 30$ Hz; Layer 11 selects the optimal f_{SWV} given the measured k^0_{ET}.

The critical design implication: $k^0_{\mathrm{ET,eff}}$ spans four orders of magnitude across the plausible tether-distance range. The SWV parameters (frequency, amplitude) cannot be fixed without first measuring k^0_{ET} on the fabricated electrode.

X. Limitations

1. **k^0_{ET} must be measured empirically.** All values in Table 3 are model-derived estimates, not measurements. The standard approach is to measure k^0_{ET} from the shift in SWV peak position as a function of f_{SWV} (Laviron's method) or from the f-dependence of the AC impedance. Until k^0_{ET} is measured on the fabricated electrode, the SWV operating parameters (Layer 11) remain uncalibrated.

2. **β for the PEG-peptide tether is unconfirmed.** Smalley 1995 [2] reports $\beta = 0.95$ Å^{-1} for *alkyl* chains. PEG tethers and flexible peptide chains have a different electronic structure; literature values range from $\beta \approx 0.6$ Å^{-1} (through-bond PEG) to $\beta \approx 1.1$ Å^{-1} (alkyl). This introduces a factor of $\exp[0.5 \times 15] \approx 10^3$ uncertainty in k^0_{ET} for $d = 15$ Å.

3. **λ_{MB} is unconfirmed.** The value $\lambda = 0.85$ eV is taken from Chidsey 1991 [3] for the Fc/alkyl/Au system. MB has a different inner-sphere reorganisation energy (λ_i) that has not been measured in any paper in `papers/layer_10/`. A 0.1 eV uncertainty in λ propagates to $\approx 20\%$ uncertainty in k^0_{ET} through $\exp(-\Delta\lambda/4k_{\mathrm{B}}T)$.

4. **The formal potential $E^{\circ\prime}_{\mathrm{Fc}}$ is not confirmed from papers in this folder.** Section VI derives $E^{\circ\prime}_{\mathrm{MB}}(\mathrm{pH})$; the corresponding Fc formal potential must be measured on the fabricated chip. Literature reports range from +0.40 to +0.47 V vs Ag/AgCl (sat.), depending on electrolyte and electrode geometry.

References

All papers in `electrochemical_device/papers/layer_10/`.

[1] Marcus RA (1965) On the theory of electron-transfer reactions. VI. Unified treatment for homogeneous and electrode reactions. *J Chem Phys* 43:679–701. DOI: 10.1063/1.1696792. **marcus1965.pdf** – Eq. 82 (electrochemical ΔF^*), Eqs. 31/33 (rate constant), Eq. 83 ($\lambda = \lambda_i + \lambda_o$), Eq. 90 ($\lambda_o$ continuum), Eq. 87b (α) confirmed.

[2] Smalley JF, Feldberg SW, Chidsey CED, Linford MR, Newton MD, Liu Y-P (1995) The kinetics of electron transfer through ferrocene-terminated alkanethiol monolayers on gold. *J Phys Chem* 99:13141–13149. DOI: 10.1021/j100035a016. **smalley1995.pdf** – $\beta_n = 1.21 \pm 0.05$ $\mathrm{CH_2^{-1}}$, $k_{T,s,n=0} = 6 \times 10^8$ s^{-1} confirmed from abstract and text.

[3] Chidsey CED (1991) Free energy and temperature dependence of electron transfer at the metal-electrolyte interface. *Science* 251:919–922. DOI: 10.1126/science.251.4996.919. **chidsey1991.pdf** – $\lambda = 0.85$ eV, $\nu_p = 6.73 \times 10^4$ $\mathrm{s}^{-1}\mathrm{eV}^{-1}$, $k^0 = 1.3$ s^{-1} (C16), BV Eq. 3, Marcus-Chidsey integral Eqs. 7–8 confirmed.

[4] Pheeney CG, Barton JK (2012) DNA electrochemistry with tethered methylene blue. *Langmuir* 28:7063–7070. DOI: 10.1021/la300566x. **pheeney2012.pdf** – $E^{\circ\prime}_{\mathrm{MB}} = -290$ mV vs Ag/AgCl at pH 7, $n = 2$, surface coverage 1.3–3.2 pmol/cm^2 confirmed from text (p. 7065).

[5] Becka AM, Miller CJ (1993) Electrochemistry at ω-hydroxy thiol coated electrodes. 4. Comparison of the double layer at ω-hydroxy thiol and alkanethiol monolayer coated Au electrodes. *J Phys Chem* 97:6233–6239. DOI: 10.1021/j100125a020. **becka1993.pdf** – Frumkin correction Eq. 5, two-capacitor model Eqs. 1–4, $C_{gc} = 238.2 c^{1/2}$ μF cm^{-2} (Eq. 7) confirmed.